

## BRAF V600K vs. BRAF V600E: a comparison of clinical and dermoscopic characteristics and response to immunotherapies and targeted therapies.

PMID: 35080260 | Indexed in PubMed

A number of mutations related to malignant melanoma (MM) have been identified, and of the mutated genes, BRAF has been found to be altered in > 50% of cases. Most of these have been BRAF V600E mutations, whereas the incidence of BRAF V600K may vary from 10% to 30%. Little is known about the clinical prognostic correlations of BRAF V600K MMs. We evaluated the clinical and dermoscopic features, incidence, therapy response and outcomes in the medium to long term. To compare the clinical and dermoscopic characteristics, the response to systemic therapies and the prognosis among MMs with BRAF V600E and BRAF V600K mutations. We retrieved the data of patients tested in our centre for MM from 2012 to 2015, including clinical features, dermoscopic pictures, clinical history and tumour mutations. Only patients with BRAF V600E and BRAF V600K mutations were included. Any MMs positive for BRAF V600K mutation were collected, and the number of V600K cases and their features were used to extract the same number of patients with BRAF V600E from our database using a matching method. The clinical and dermoscopic presentation, therapy response and disease progression of the two groups were then evaluated. In total, 132 cases of BRAF V600E-mutated MMs were identified, and then randomized with a propensity-score method to match the 10 retrieved cases of BRAF V600K mutation. Both groups had a nodular appearance to the tumours and an advanced disease stage, and no significant differences in dermoscopic features were highlighted. During the follow-up period, four patients with BRAF V600K died of disease-specific causes. Moreover, we found a higher frequency of metastasis, a faster disease progression and more rapid mortality in patients with BRAF V600K. Despite the small size of this study, the results show similar clinical and dermoscopic characteristics between V600E and V600K mutations, but compared with BRAF V600E MMs, BRAF V600K MMs seem to be less responsive to therapy and have a worse prognosis.

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## Dabrafenib in the treatment of advanced melanoma.

PMID: 23807941 | Indexed in PubMed

Advanced melanoma has long been a challenging malignancy to treat due to a relative paucity of efficacious therapeutic options. However, the identification of activating BRAF mutations in approximately 50% of patients with cutaneous melanoma has ushered in the era of targeted therapy for melanoma patients. Similar to the first-in-class selective serine/threonine-protein kinase B-raf inhibitor vemurafenib, dabrafenib is highly efficacious in melanoma patients with BRAF V600E mutations, with response rates of approximately 50% and progression-free survival of 6 months. There is data to suggest that dabrafenib not only shows activity in V600E-mutated melanoma, but also in non-V600E BRAF-mutated disease such as V600K. There is also early data to suggest that dabrafenib is effective in controlling metastases in the brain. Combining dabrafenib with the selective mitogen-activated protein kinase kinase (MEK) inhibitor trametinib has been effective in improving both the progression-free survival and overall survival of melanoma patients over those patients treated with dabrafenib alone. Dabrafenib is still being evaluated in several clinical trials in melanoma as well as a variety of other solid tumors with BRAF mutations. The U.S. Food and Drug Administration has recently approved dabrafenib as a single agent for the treatment of unresectable or metastatic melanoma in adult patients with BRAF V600E mutation.

## Identification of aurora kinase B and Wee1-like protein kinase as downstream targets of (V600E)B-RAF in melanoma.

PMID: 23416158 | Indexed in PubMed

BRAF is the most mutated gene in melanoma, with approximately 50% of patients containing V600E mutant protein. (V600E)B-RAF can be targeted using pharmacological agents, but resistance develops in patients by activating other proteins in the signaling pathway. Identifying downstream members in this signaling cascade is important to design strategies to avoid the development of resistance. Unfortunately, downstream proteins remain to be identified and therapeutic potential requires validation. A kinase screen was undertaken to identify downstream targets in the (V600E)B-RAF signaling cascade. Involvement of aurora kinase B (AURKB) and Wee1-like protein kinase (WEE1) as downstream proteins in the (V600E)B-RAF pathway was validated in xenografted tumors, and mechanisms of action were characterized in size- and time-matched tumors. Levels of only AURKB and WEE1 decreased in melanoma cells, when (V600E)B-RAF, mitogen-activated protein kinase 1/2, or extracellular signal-regulated kinase 1/2 protein levels were reduced using siRNA compared with other identified kinases. AURKB and WEE1 were expressed in tumors of patients with melanoma at higher levels than observed in normal human melanocytes. Targeting these proteins reduced tumor development by approximately 70%, similar to that observed when inhibiting (V600E)B-RAF. Furthermore, protein or activity levels of AURKB and WEE1 decreased in melanoma cells when pharmacological agents targeting upstream (V600E)B-RAF or mitogen-activated protein kinase were used to inhibit the (V600E)B-RAF pathway. Thus, AURKB and WEE1 are targets and biomarkers of therapeutic efficacy, lying downstream of (V600E)B-RAF in melanomas.

## The clinical response to vemurafenib in a patient with a rare BRAFV600DK601del mutation-positive melanoma.

PMID: 25265970 | Indexed in PubMed

Mutations in the activation segment of the v-raf murine sarcoma viral oncogene homolog B (BRAF) gene are present in approximately 50% of melanomas. The selective BRAF inhibitor vemurafenib has demonstrated significant clinical benefits in patients with melanomas harboring the most common mutations (V600E, V600K and V600R). However, the clinical activity of BRAF inhibitors in patients with rare mutations of codon 600 and the surrounding codons has not been documented. We used the BRAF inhibitor vemurafenib to treat a patient presenting a rare p.V600\_K601delinsD-mutated melanoma. An objective response was evidenced by two months of progression-free survival. By cloning and sequencing BRAF exon 15, we confirmed that a dual mutation was present on a single allele and thus resulted in a BRAFV(600DK601del) mutant protein. We also performed an in silico crystal structure analysis of the mutated protein, in order to characterize the nature of the putative interaction between vemurafenib and the mutant protein. This clinical experience suggests that (i) patients with BRAFV(600DK601del)-mutation-positive melanoma can be treated successfully with the oral BRAF inhibitor vemurafenib and (ii) molecular screening in this context should encompass rare and complex mutations.

## Successful rechallenge in two patients with BRAF-V600-mutant melanoma who

experienced previous progression during treatment with a selective BRAF inhibitor.

PMID: 22584957 | Indexed in PubMed

The v-raf murine sarcoma viral oncogene homolog B1 (BRAF) gene is mutated at position 600 in about 50% of melanoma. Mutant BRAF activates the downstream effectors of the RAS-RAF-MEK-MAPK pathways and is a driver oncogene in these melanoma cells. Selective BRAF-V600 inhibitors (vemurafenib, dabrafenib) have high antitumor activity against BRAF-V600-mutant melanoma with objective tumor response rates. Resistance, however, develops within less than a year in the majority of patients. Several different mechanisms have been found to mediate acquired resistance, but these do not involve the occurrence of secondary mutations in the BRAF gene. Two patients with BRAF-V600E mutant melanoma who had documented progression during treatment with dabrafenib/GSK1120212 and dabrafenib, respectively, were rechallenged with dabrafenib and vemurafenib after a treatment-free interval of 8 and 4 months during which further progression was documented. Both patients showed a marked clinical response and, in both, objective tumor regression (qualifying as a mixed and a partial response according to RECIST) was documented. These two case observations indicate that resistance to BRAF-selective inhibitors can be reversible following treatment interruption.

Use of BRAF immunohistochemistry as a screening test in detecting BRAFV600E mutation in melanomas.

PMID: 34018988 | Indexed in PubMed

BRAF mutation is detected in 50-70% of melanomas. The molecular methods used to detect BRAF mutations are 80-90% sensitive, specific, and expensive methods. Immunohistochemistry is a relatively common, rapid, relatively inexpensive method in pathology practice compared to molecular techniques. We aimed to compare immunohistochemical and molecular methods in our case of malign melanoma in which we investigated BRAF mutation with "real time PCR" method and to investigate the compatibility of molecular test results of BRAF immunohistochemistry results as a preliminary test. Selected blocks of 30 patients with metastatic melanoma who came to our department for BRAF mutation detection were subjected to real time PCR molecular method and immunohistochemical study was performed with BRAF primer antibody. BRAF mutation was detected by molecular method in 7 of 30 cases (23.33%). In all of these 7 cases, positive immunohistochemical staining was identified (100%). In conclusion, the use of BRAF immunohistochemistry as a screening test in the detection of mutant disease will allow the cost-effective use of molecular testing.

BRAF mutation correlates with worse local-regional control following radiation therapy in patients with stage III melanoma.

PMID: 34537078 | Indexed in PubMed

In patients with stage III melanoma, the use of adjuvant radiation therapy (RT) after lymph node dissection (LND) may be currently considered in selected high-risk patients to improve tumor control. Melanomas harbor BRAF mutations (BRAF+) in 40-50% of cases, the majority of which are on the V600E residue. This study sought to compare the clinical outcomes after RT between patients with BRAF+ and BRAF- melanoma. This was a retrospective review of 105 Stage III melanoma patients treated at our

institution with LND followed by adjuvant RT from 2006 to 2019. BRAF mutational status was determined on the primary skin or nodal tissue samples from all patients. We compared characteristics of the BRAF+ and BRAF- groups using Fisher's exact test and Wilcoxon rank sum test and performed univariate and multivariate analysis using Kaplan-Meier estimates, log-rank tests, and Cox proportional hazards modeling with the clinical outcomes of local-regional lymph node control, distant metastasis-free survival (DMFS), recurrence-free survival (RFS), and overall survival (OS). Fifty-three (50%) patients harbored a BRAF mutation (92%, pV600E). BRAF+ patients were younger and had primary tumors more commonly found in the trunk vs head and neck compared to BRAF- patients (p 4 times increased risk of local recurrence or regional lymph node recurrence). These results could be useful for adjuvant RT consideration in lymph node positive melanoma patients and supports other data that BRAF mutation confers radiation resistance.

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## Plasma proteome alterations by MAPK inhibitors in BRAFV600-mutated metastatic cutaneous melanoma.

PMID: 34246984 | Indexed in PubMed

Approximately half of metastatic cutaneous melanomas (CM) harbor a mutation in the BRAF protooncogene, upregulating the mitogen-activated protein kinase (MAPK)-pathway. The development of inhibitors targeting the MAPK pathway (MAPKi), i.e., BRAF- and MEK-inhibitors (BRAFi and MEKi), have substantially improved the survival in BRAFV600E/K-mutated stage IV metastatic CM. However, most patients develop resistance to treatment and no predictive biomarkers exist in practice. This study aimed at discovering plasma proteome changes during treatment MAPKi in patients with metastatic (stage IV) CM. Matched plasma samples before (pre) and during treatment (trm) from 23 patients with stage IV CM, treated with BRAF-inhibitors (BRAFi) alone or BRAF- and MEK- inhibitors combined (BRAFi and MEKi), were collected and analyzed with targeted proteomics by proximity extension assays. Additionally, plasma from 9 patients treated with BRAFi and MEKi was analyzed with in-depth high-resolution isoelectric focusing liquid-chromatography mass-spectrometry proteomics. Alterations of plasma proteins involved in granzyme and interferon gamma pathways were detected in patients treated with BRAFi, and cell adhesion-, neutrophil degranulation-, and proteolysis pathways in patients treated with BRAFi and MEKi. Several proteins were associated with progression-free survival after MAPKi treatment. We show that the majority of the altered plasma proteins were traceable to BRAFV600E-mutant metastatic CM tissue at mRNA level in 154 patients from the TCGA, further strengthening their involvement in tumoral response to treatment. This wide screen of plasma proteins unravels proteins that may serve as predictive and/or prognostic biomarkers of MAPKi treatment, opening a window of opportunity for plasma biomarker discovery in MAPKi-treatment of BRAFV600-mutant metastatic CM.

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## Two Case Reports of Rare BRAF Mutations in Exon 11 and Exon 15 with Discussion of Potential Treatment Options.

PMID: 27790118 | Indexed in PubMed

BRAF mutations occur in up to 50% of melanomas. Mutations in the BRAF gene directly influence the patient's treatment because several inhibitors are available that only target BRAFV600 mutations. Herein, we describe two cases of patients with metastatic melanomas, each carrying a 'nonstandard'

mutation in the BRAF gene: BRAFK601E and BRAFG466E, respectively. The first patient was treated with a MEK inhibitor and the second one with ipilimumab. However, not all BRAF mutations result in increased BRAF kinase activity, and clinical data for 'nonstandard' mutations, such as those described in our case report, are sparse. Therefore, treatment with MEK inhibitors can be helpful in cases where BRAF mutations result in increased activity, whereas immune checkpoint inhibitors might be used in cases where the mutations lead to activity levels below those of the wild type.

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## BRAF mutation in 'sarcomas': a possible method to detect de-differentiated melanomas.

PMID: 24117833 | Indexed in PubMed

BRAF is mutated in 50-60% of melanomas, but BRAF mutation in sarcomas has not been systematically evaluated. Some melanomas are spindle and may show no immunohistochemical evidence of melanocytic differentiation. Similarly, many sarcomas are undifferentiated, i.e. undifferentiated pleomorphic sarcomas (UPS). Diagnosing melanoma versus sarcoma in an undifferentiated spindle cell malignancy can be challenging. Our aim was to evaluate the prevalence of BRAF mutation in sarcomas and the use of BRAF mutational status in the diagnosis of spindle cell malignancies. BRAF mutational analysis was performed on tissue from 104 patients: 90 with sarcoma only (50 UPS) and 14 with sarcoma and melanoma (seven UPS). In the sarcoma-only group, BRAF mutation was absent. In the sarcoma-melanoma group, three sarcomas showed BRAF mutation; all were UPS, occurred after the melanomas and did not stain for melanocytic markers. One melanoma-sarcoma pair showed identical BRAF V600E mutations. The presence of BRAF mutation in these tumours raises the possibility that poorly differentiated spindle cell malignancies with BRAF mutation may represent melanomas, and BRAF mutational analysis should be considered in a patient with a spindle cell malignancy and a history of melanoma, as a positive result may indicate de-differentiated melanoma.

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